APPLIED MACHINE LEARNING · ASSIGNMENT 2

Transmission-Line Fault Detection Using Logistic Regression and SVM

Can a model tell a fault from normal operation using Ia, Ib, Ic, Va, Vb, Vc? · Laiba Afzaal and Muhammad Ali

The problem

A protection relay must trip the breaker on a fault and stay quiet otherwise. A **missed fault** damages equipment. A **false alarm** causes an avoidable outage.

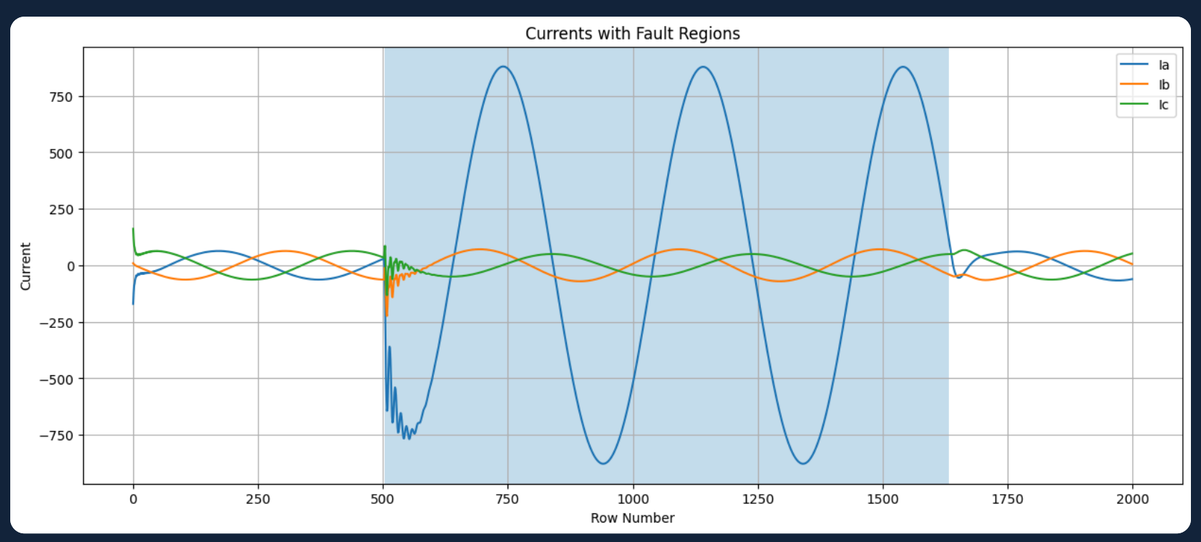
12,001 samples, 6 features

5,496 fault rows

6,505 no-fault rows

Random 60/20/20 split (seed 42): 7,200 train · 2,400 validation · 2,401 test. All features standardised with training statistics only.

What a fault looks like



Real data, first 2,000 rows, shaded = fault. Ia swings from about ±60 to about ±880. The wave crosses zero either way, so the **sign** of one sample says nothing. The **size** does.

Squares fix it

Logistic regression (written by hand) and a linear SVM both stall near **73%**: two algorithms, one ceiling. The classes are not linearly separable. Adding squared features lifts logistic regression to **97.9%**.

$I^2 > 0$

+18.9 · +4.8 · +14.0
current surges in a fault

$V^2 < 0$

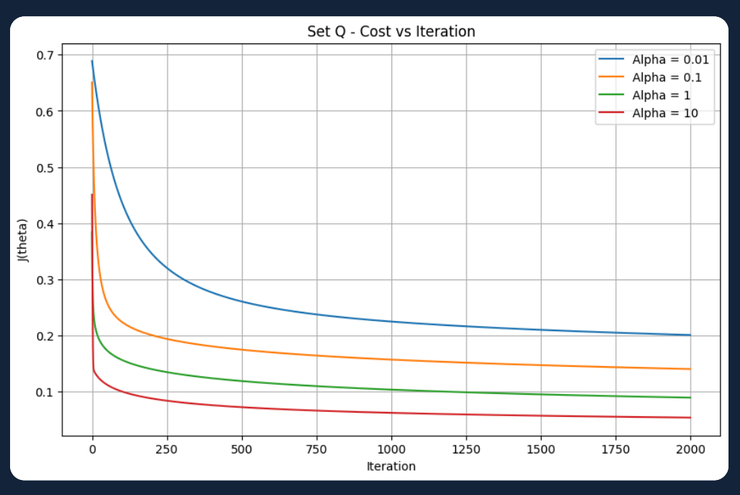
−6.0 · −6.5 · −6.1
voltage collapses in a fault

Final test results · 2,401 rows, used once

Model	Accuracy	TN	False alarms	Missed faults	TP
Logistic regression, Set R	72.64%	1305	0	657	439
Logistic regression, Set Q	97.88%	1299	6	45	1051
Linear SVM (C=100)	73.64%	1305	0	633	463
RBF SVM (C=100, γ=1)	**99.58%**	**1301**	**4**	**6**	**1090**

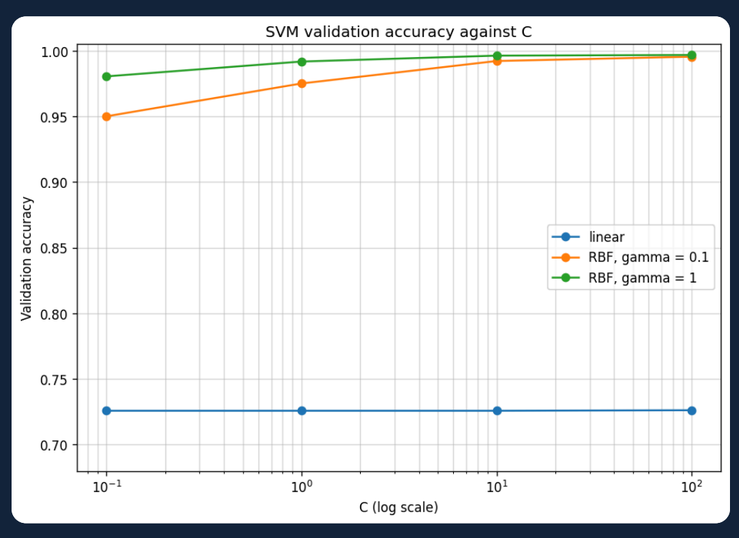
False alarm = unneeded trip. Missed fault = breaker fails to trip, the dangerous error. Of 1,096 real faults: RBF SVM misses 6, Set Q misses 45, linear models miss about 630 to 660.

Cost J(θ), Set Q



Every α falls at every iteration; α = 10 is the largest valid step (J = 0.053). On raw features (Set R) α = 10 diverged.

Validation accuracy vs C



RBF climbs to 99.7% as C grows. The linear SVM is flat near 73% for every C.

Missed faults (of 1,096)

Logistic, Set R — 657 · 59.9%

Linear SVM — 633 · 57.8%

Logistic, Set Q — 45 · 4.1%

RBF SVM — 6 · 0.55%

Bars are scaled to the worst model. Lower is safer.

Support vectors

5,835 vs 266

Linear vs RBF, of 7,200 training rows. 81% support vectors means no clean gap: the model is too simple.

What C does

848 → 266

RBF support vectors (γ=1) as C goes 0.1 → 100, while accuracy rises 98.0% → 99.7%. C is the price of a margin violation.

A caution

Optimistic

Consecutive samples leak near-identical neighbours into the test set. A real relay sees new faults.

Chosen settings (validation set)

Model	Setting	Val. acc.	J / SVs
LR, Set R	α = 1	73.08%	J 0.677
LR, Set Q	α = 10	97.83%	J 0.053
Linear SVM	C = 100	72.62%	5,835
RBF SVM	C=100, γ=1	99.67%	266

α = largest value for which J falls at every iteration. C and γ chosen on validation accuracy. J / SVs = final training cost (LR) or support-vector count (SVM).

VERDICT

Deploy the RBF SVM: 6 missed faults, 4 false alarms.

Judge protection by missed faults, not accuracy. Next: re-test on a time-block split, bias the threshold toward "fault", and trip only after several consecutive fault samples.

Code, notebook and web demo: [GitHub link]

Full write-up: [Medium link]